Numerical Methods

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Syllabus (part of PHSDSC405P: Mathematical Methods II Lab)

Reference
J. Mathews and R. L. Walker, *Mathematical Methods of Physics*

I. INTERPOLATION

Suppose we have a two-column data set with equally spaced x values, where $x_i = x_0 + ih$.

x_0	x_1	x_2	$\cdots$	x_{n-2}	x_{n-1}	x_n
y_0	y_1	y_2	$\cdots$	y_{n-2}	y_{n-1}	y_n

A. Forward Differences

We define the difference operator Δ such that,

$$\Delta y_i = y_{i+1} - y_i \Rightarrow y_{i+1} = (1+\Delta) y_i \Rightarrow y_i = (1+\Delta)^i y_0, \tag{1}$$

which we can expand using,

$$\begin{aligned} \Delta^2 y_i &= \Delta(\Delta y_i) = \Delta y_{i+1} - \Delta y_i, \ldots, \\ \Delta^n y_i &= \Delta(\Delta^{n-1} y_i) = \Delta^{n-1} y_{i+1} - \Delta^{n-1} y_i. \end{aligned} \tag{2}$$

To interpolate the value of y at some intermediate value x, such that $x_0 < x < x_n$, we use the same formula as the one used for points where data is available. I.e.,

$$y(x) = (1+\Delta)^\alpha y_0, \tag{3}$$

where, $x = x_0 + \alpha h \Rightarrow \alpha = \dfrac{x - x_0}{h}$. Expanding, we get,

$$y(x) = y_0 + \alpha \Delta y_0 + \frac{\alpha(\alpha-1)}{2!}\Delta^2 y_0 + \cdots \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}\Delta^n y_0. \tag{4}$$

Note that even for non-integer α the series terminates since differences of the order greater than n are undefined for a data set with n entries. The above relation also simplifies to,

$$y(x) = y_0 + \frac{(x-x_0)}{h}\Delta y_0 + \frac{(x-x_0)(x-x_1)}{2! h^2}\Delta^2 y_0 + \cdots \frac{(x-x_0)(x-x_1)\cdots(x-x_{n-1})}{n! h^n}\Delta^n y_0. \tag{5}$$

This is the Newton-Gregory forward difference interpolation formula.
When using this formula using pen and paper, it is useful to construct a table similar to the one shown below (5 data points are used for illustration).

x_0	y_0				
		Δy_0			
x_1	y_1		$\Delta^2 y_0$		
		Δy_1		$\Delta^3 y_0$	
x_2	y_2		$\Delta^2 y_1$		$\Delta^4 y_0$
		Δy_2		$\Delta^3 y_1$	
x_3	y_3		$\Delta^2 y_2$		
		Δy_3			
x_4	y_4				

TABLE I: Sample forward difference table for a data set with five entries.

B. Backward Differences

In the same way as above, we may define the backward difference operator ∇ such that,

$$\nabla y_i = y_i - y_{i-1} \Rightarrow y_{i-1} = (1-\nabla)y_i \Rightarrow y_{n-i} = (1-\nabla)^i y_n, \tag{6}$$

which we can expand using,

$$\begin{aligned} \nabla^2 y_i &= \nabla(\nabla y_i) = \nabla y_i - \nabla y_{i-1}, \ldots, \\ \nabla^n y_i &= \nabla(\nabla^{n-1} y_i) = \nabla^{n-1} y_i - \nabla^{n-1} y_{i-1}. \end{aligned} \tag{7}$$

To interpolate the value of y at some intermediate value x, such that $x_0 < x < x_n$, we use the same formula as the one used for points where data is available. We write $x = x_n + \alpha h \Rightarrow \alpha = \dfrac{x - x_n}{h}$. Thus,

$$y(x) = (1-\nabla)^{-\alpha} y_n. \tag{8}$$

Expanding, we get,

$$y(x) = y_n + \alpha \nabla y_n + \frac{\alpha(\alpha+1)}{2!}\nabla^2 y_n + \cdots \frac{\alpha(\alpha+1)\cdots(\alpha+n-1)}{n!}\nabla^n y_0. \tag{9}$$

As before, the above relation also simplifies to,

$$y(x) = y_n + \frac{(x-x_n)}{h}\nabla y_n + \frac{(x-x_n)(x-x_{n-1})}{2!h^2}\nabla^2 y_n + \cdots \frac{(x-x_n)(x-x_{n-1})\cdots(x-x_1)}{n!h^n}\nabla^n y_n. \tag{10}$$

This is the Newton-Gregory backward difference interpolation formula.

x_0	y_0				
		∇y_1			
x_1	y_1		$\nabla^2 y_3$		
		∇y_2		$\nabla^3 y_3$	
x_2	y_2		$\nabla^2 y_3$		$\nabla^4 y_4$
		∇y_3		$\nabla^3 y_4$	
x_3	y_3		$\nabla^2 y_4$		
		∇y_4			
x_4	y_4				

TABLE II: Sample backward difference table for a data set with five entries.